Yiyang (Ian) Chen

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EDUCATION

University of California, Los Angeles **Sep 2023 – Jun 2026**
B.S. Cognitive Science · GPA 3.91 · Dean's Honors List

The Barstow School – Ningbo Campus **Sep 2019 – Jun 2023**
High School Diploma · GPA 4.51 · High Honor Roll

RESEARCH EXPERIENCE

Luo Lab, Department of Human Genetics, UCLA **Nov 2023 – Present**
Independent Researcher · Advisor: Chongyuan Luo, Ph.D.

- Developed new in-situ DNA barcoding methods to map neuronal cell types and gene regulatory states with significantly increased spatial specificity (from ~366 µm spatial grids to adaptive ~30 µm single-cell bins) and throughput to cover full mouse brain cross-sections (~1.2 cm^2).
- Designed and built custom optofluidic microscope systems for photonic-indexing experiments, including full microfluidic systems, hardware-software interfaces, and end-to-end software for automated fluorescence imaging, reagent flow, patterned laser excitation, and data collection.
- Independently wrote and published spatial combinatorial barcode design tools as open-source software (github.com/yiyangc22/Mercury-Redstone, DOI: 10.5281/zenodo.22650611).
- Proposed and designed statistical models to decode single-cell spatial barcodes from high-noise sequencing reads, significantly increasing readout confidence using ambient decontamination.
- Cross-validated mouse brain neuron cell-type distribution, methylation dynamics, and regional identity by integrating single-nucleus methylome data against reference brain atlases.
- Independently proposed, designed, and published a universal single-cell/single-molecule spatial mapping and visualization tool for photonic-indexing barcode experiments and spatial-temporal migratory stream analyses (github.com/yiyangc22/Maplet, DOI: 10.5281/zenodo.22651477).
- Developed machine-vision cell segmentation tool with human-in-the-loop training to create full-tissue cell database, reducing processing time from ~30 s to ~2.5 s per image at >90% accuracy.

Golshani Lab, Department of Neurology, UCLA **Apr 2026 – Present**
Student Research Assistant · Advisor: Peyman Golshani, M.D., Ph.D.

- Developed custom machine vision software to identify and extract neuronal firing patterns in low signal-to-noise ratio voltage imaging recordings of mouse hippocampal CA1 region activity.
- Developed a reliable, frame-consistent high-speed image recording hardware-software interface for calcium and voltage imaging systems using a multi-process pulse-burst acquisition method.
- Designed software programs to enhance continuous-acquisition mouse behavior recording with real-time quantification and pattern analysis during behavior training and learning tasks.

Shanghai Xuhui District Central Hospital **May 2021 – Sep 2021**
Synthetic Biology Summer Research Intern

- Engineered and deployed synthetic plasmids into gram-negative bacteria to develop a dose-gated growth-inhibition mechanism. Conducted under an iGEM silver medal project in 2021.

PUBLICATIONS & PRESENTATIONS

Nguyen R., **Chen Y.**, Galasso J., Zhang Y., Williams M., Kakhandiki S., Perillo E., Parsaee E., Tremel G., Rane T., Chen P., Ecker J.R., Beechem J., Flint J., Jung J., Luo C. PSI-seq: photonic single-nucleus indexing profiling of the genome, 3D genome conformation, and epigenomes. Manuscript: *in preparation.*

Chen Y. Mercury-Redstone: a Python package for designing single-cell combinatorial photonic-indexing barcodes (Version 1.24) [Computer software]. Zenodo, Sep 2026, https://doi.org/10.5281/zenodo.22650612.

Chen Y. Maplet: a local, web-based viewer for spatial single-cell and single-molecule datasets (Version 0.11) [Computer software]. Zenodo, Sep 2026, https://doi.org/10.5281/zenodo.22738657.

Chen Y., Luo C. Mapping neuronal migration in developing mammalian brains with photonic-indexing sequencing (Pi-seq), UCLA Undergraduate Research Showcase 2025, Los Angeles, CA. Conference presentation: May 2025.

AWARDS & HONORS

Certificate of Recognition for Outstanding Commitment, UCLA Health Volunteer Services 2026
Dean's Honors List, UCLA 2024
Principal's List with High Honor Roll, Barstow School 2023
Mary Barstow Award with Hanvos Kent First Prize Scholarship (~$17,500), Barstow School 2022
Group Silver Medal, iGEM Competition 2021
Co-inventor, Chinese utility model patent 2019215924710 2020

ADDITIONAL EXPERIENCE

DataAnnotation **Jun 2026 – Nov 2026**

AI Language Model Training & Bilingual Expert

- Trained large language models on multilingual natural language processing, specializing in reinforcement learning from human feedback (RLHF) on multilingual input/output. Part-time.

UCLA Health **Apr 2024 – Jun 2026**

Non-Clinical Undergraduate Health Sciences Volunteer

- Designed an automated file transfer system across UCLA's Hoffman2 compute cluster and Box data management platform, achieving near-instant read-write across multiple devices.
- Created and presented scientific figures, posters, and slide presentations for research labs at UCLA's David Geffen School of Medicine and Department of Human Genetics.

Ningbo Yuxun Information Technology **Mar 2019 – Feb 2020**

Software Development Intern

- Co-wrote automated RFID processing and database communication software (Java, SQL) for a product traceability system. Co-inventor on Chinese utility model patent 2019215924710.

TECHNICAL SKILLS

Programming – Python (PyTorch, cellpose, scanpy, ALLCools, harmony, PyQt5, Matplotlib, Plotly, Seaborn), R (RStudio, Quarto), LabVIEW, MATLAB, Java, C#, C++, C, JavaScript/TypeScript, SQL, etc.
Single-cell analysis – DNA CH/CG methylation analysis, harmony integration, cell type labeling, Leiden clustering, KNN label transfer, dimensionality reduction and UMAP visualization, etc.
Instrumentation – Hardware-software interface design, custom microscopy, image acquisition, fluidics, and patterned laser illumination systems, computer-aided design (CAD) and modeling, etc.
Computation – Computer vision for cell biology, statistical analysis and model fitting, machine learning and model training for biological data, data structure optimization and design, etc.
Wet-bench – Fluorescence microscopy, PCR, CRISPR, gel electrophoresis, spectrophotometry, etc.

LANGUAGES

English (native-level), Mandarin Chinese (native)